BIO503 – BIOLOGICAL PHYSICS

FINAL TERM IMPORTANT QUESTIONS

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Admin: **BS Biological Sciences** (One place to all your solutions)

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SHORT QUESTIONS

1. Write applications of NMR Spectroscopy.

Applications of NMR Spectroscopy

Nuclear Magnetic Resonance (NMR) spectroscopy is an analytical technique used to determine the structure and composition of molecules. Its important applications are:

- ***Structure Determination:*** Helps determine the molecular structure and arrangement of atoms in organic compounds.
- ***Identification of Compounds:*** Used to identify unknown compounds by analyzing their NMR spectra.
- ***Purity Testing:*** Detects impurities and checks the purity of chemicals and pharmaceutical products.
- ***Functional Group Analysis:*** Identifies different functional groups & their chemical environment in a molecule.
- ***Reaction Monitoring:*** Monitors the progress of chemical reactions and helps study reaction mechanisms.
- ***Quantitative Analysis:*** Measures the concentration of compounds present in a mixture.
- ***Biological Studies:*** Used to study proteins, nucleic acids, and other biomolecules.
- ***Medical Applications:*** Forms the basis of MRI**,** which is used for non-invasive diagnosis of diseases.

2. What are good qualities of sources of energy?

Good Qualities of Sources of Energy

A good source of energy is one that provides a large amount of useful energy safely, economically, and with minimum environmental impact.

- ***High Energy Output:*** It should produce a large amount of energy per unit mass or volume.
- ***Easily Available:*** The source should be readily available and accessible.
- ***Economical:*** It should be affordable to obtain, transport, and use.
- ***Easy to Store and Transport:*** The energy source should be convenient to store and move from one place to another.
- ***Safe to Use:*** It should be easy and safe to handle with low risk of accidents.
- ***Reliable:*** It should provide a continuous and dependable supply of energy.

3. Write some techniques for analysis of macromolecules.

Techniques for the Analysis of Macromolecules

Macromolecules such as proteins, nucleic acids, and polymers are analyzed using different analytical techniques to determine their structure, composition, and properties.

- ***NMR Spectroscopy:*** Used to determine the molecular structure and study the arrangement of atoms in macromolecules.
- ***Mass Spectrometry (MS):*** Determines the molecular weight and composition of macromolecules.
- ***X-ray Crystallography:*** Provides the three-dimensional structure of proteins and other large molecules.
- ***Gel Electrophoresis:*** Separates DNA, RNA, and proteins based on their size and charge.
- ***Chromatography:*** Used to separate, purify, and identify macromolecules.
- ***UV–Visible Spectroscopy:*** Measures the concentration and purity of proteins and nucleic acids.
- ***Infrared (IR) Spectroscopy:*** Identifies functional groups and chemical bonds present in macromolecules.

4. Write some examples of friction in daily life.

Examples of Friction in Daily Life

Friction is the force that opposes the motion of one surface over another. It plays an important role in many everyday activities.

- ***Walking:*** Friction between our feet and the ground prevents slipping.
- ***Writing:*** Friction between a pen or pencil and paper allows us to write.
- ***Braking a Vehicle:*** Friction between the brake pads and wheels slows down or stops the vehicle.
- ***Driving:*** Friction between the tyres and the road provides grip and prevents skidding.
- ***Lighting a Matchstick:*** Friction produces heat, which ignites the match.
- ***Holding Objects:*** Friction between our hands and an object helps us grip it firmly.
- ***Climbing Stairs:*** Friction between shoes and the steps prevents slipping.
- ***Erasing with an Eraser:*** Friction between the eraser and paper removes pencil marks.

5. Explain structure of amino acids.

Structure of Amino Acids

Amino acids are the basic units of proteins. They are organic compounds containing an amino group ($–NH_2$) and a carboxyl group (–COOH).

- A typical amino acid consists of a central carbon atom (α-carbon) attached to four different groups:
    - **Amino group ($–NH_2$)**
    - **Carboxyl group (–COOH)**
    - **Hydrogen atom (–H)**
    - **Side chain (R group)**

- The R group varies among different amino acids and determines their properties and functions.
- General structure of an amino acid:
    - **NH_2 – CH(R) – COOH**
- Amino acids join together through peptide bonds to form proteins.

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6. State types of Self Assembly.

Types of Self-Assembly

Self-assembly is the process in which molecules or particles spontaneously organize themselves into ordered structures without external control.

- ***Molecular Self-Assembly:*** Formation of organized structures through interactions between individual molecules.
- ***Supramolecular Self-Assembly:*** Assembly of molecules through non-covalent interactions such as hydrogen bonding, electrostatic forces, and van der Waals forces.
- ***Colloidal Self-Assembly:*** Organization of small particles suspended in a liquid to form ordered structures.
- ***Biological Self-Assembly:*** Natural formation of structures such as proteins, DNA, and cell membranes.
- ***Surface Self-Assembly:*** Arrangement of molecules or particles on solid surfaces to form organized layers.

7. Write four types of friction.

Types of Friction

Friction is the force that opposes the relative motion between two surfaces in contact. The four main types of friction are:

- ***Static Friction:*** The friction that prevents an object from starting to move when a force is applied.
- ***Sliding Friction:*** The friction that occurs when one surface slides over another surface.
- ***Rolling Friction:*** The friction that occurs when an object rolls over a surface.
- ***Fluid Friction:*** The friction experienced by an object moving through a fluid such as air or water.

8. What is protein fractionation?

Protein Fractionation

Protein fractionation is a process used to separate a mixture of proteins into different fractions based on their physical and chemical properties such as size, charge, solubility, and molecular weight.

- It helps in the isolation and purification of specific proteins from complex biological mixtures.
- Different techniques are used for protein fractionation, including precipitation, centrifugation, chromatography, and electrophoresis.
- It is widely used in biochemistry, molecular biology, and pharmaceutical research for studying protein structure and function.
- Protein fractionation helps in the identification, characterization, and analysis of individual proteins.

9. Define enzymes.

Enzymes

Enzymes are biological catalysts that accelerate the rate of biochemical reactions in living organisms without undergoing any permanent change or being consumed during the reaction. Most enzymes are proteins made up of amino acids, although some RNA molecules can also show enzymatic activity. They work by lowering the activation energy required for a reaction and are highly specific for their substrates. Enzymes play important roles in many biological processes, including digestion, metabolism, energy production, and regulation of cellular activities.

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10. Write applications of biophysics in biology.

Applications of Biophysics in Biology

Biophysics is the branch of science that applies the principles of physics to understand biological structures and processes. Its important applications in biology are:

- ***Study of Biomolecules:*** Helps in understanding the structure and functions of proteins, DNA, RNA, and other biological molecules.
- ***Medical Imaging:*** Techniques such as MRI and ultrasound are based on biophysical principles and are used for disease diagnosis.
- ***Microscopy:*** Used to study cells and cellular structures using advanced techniques like electron microscopy.
- ***Drug Development:*** Helps in understanding drug–target interactions and designing new medicines.
- ***Molecular Biology:*** Used to study processes such as DNA replication, protein folding, and enzyme activity.
- ***Neurobiology:*** Helps in studying nerve impulses, brain functions, and electrical activities of neurons.
- ***Biomechanics:*** Applies physical principles to understand movement, muscles, bones, and body functions.

11. Define Ion-dipole force with example.

Ion-Dipole Force

An ion-dipole force is an attractive force that occurs between an ion and a polar molecule. It is caused by the electrostatic attraction between the charge of an ion and the partial charges present on a polar molecule. This type of force is important in the dissolution of ionic compounds in polar solvents.

Example:

When sodium chloride (NaCl) dissolves in water, the Na^+ ions are attracted to the oxygen end (partial negative charge) of water molecules, while Cl^- ions are attracted to the hydrogen end (partial positive charge) of water molecules. These attractions are examples of ion-dipole forces.

12. Write applications of mass spectrometry.

Applications of Mass Spectrometry

Mass spectrometry is an analytical technique used to identify compounds and determine their molecular mass and composition by measuring the mass-to-charge ratio of ions.

- ***Molecular Weight Determination:*** Used to measure the accurate molecular mass of compounds and biomolecules.
- ***Identification of Compounds:*** Helps identify unknown substances by analyzing their mass spectra.
- ***Protein and Peptide Analysis:*** Used to study the structure, sequence, and modifications of proteins.
- ***Drug Development:*** Helps in the discovery, identification, and quality control of pharmaceutical compounds.
- ***Chemical Analysis:*** Used to detect and analyze the composition of chemical mixtures.
- ***Environmental Analysis:*** Helps detect pollutants, toxins, and harmful chemicals in the environment.
- ***Medical Diagnosis:*** Used in clinical studies for detecting biomarkers and analyzing biological samples.
- ***Food Analysis:*** Helps identify contaminants and determine the composition of food products.

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13. What is entropy? What is its unit?

Entropy

Entropy is a thermodynamic property that measures the degree of disorder or randomness of a system. It represents the amount of energy in a system that is unavailable for doing useful work. A system with greater disorder has higher entropy, while an ordered system has lower entropy. The SI unit of entropy **is** joule per kelvin (J K^{-1}). For molar entropy, the unit is joule per mole per kelvin (J mol^{-1} K^{-1})**.**

14. Write methods to reduce friction.

Methods to Reduce Friction

Friction can be reduced by decreasing the resistance between two surfaces in contact. The common methods are:

- ***Lubrication:*** Applying lubricants such as oil or grease between surfaces reduces friction by forming a smooth layer.
- ***Polishing Surfaces:*** Making surfaces smoother decreases irregularities and reduces friction.
- ***Using Ball Bearings:*** Ball bearings convert sliding friction into rolling friction, which is much smaller.
- ***Streamlining:*** Designing objects with smooth shapes reduces fluid friction in air or water.
- ***Using Low-Friction Materials:*** Materials with low friction properties help reduce resistance between moving surfaces.
- ***Reducing Surface Contact:*** Minimizing the area of contact between moving parts can help lower friction.

15. Why the cell membrane is called a selectively permeable membrane?

Cell Membrane as a Selectively Permeable Membrane

The cell membrane is called a selectively permeable membrane because it allows only specific substances to pass through it while preventing the entry or exit of other substances. It controls the movement of materials such as water, gases, ions, nutrients, and waste products between the cell and its surroundings. This selective permeability is due to the phospholipid bilayer and embedded proteins present in the membrane. Small molecules can pass easily, while larger molecules and ions require special transport proteins. This property helps maintain the internal conditions of the cell, protects the cell, and ensures proper functioning of cellular activities.

16. Which has a higher melting point, ICl or Br_2? Give reason.

ICl (iodine monochloride) has a higher melting point than Br_2 (bromine). This is because ICl is a polar molecule due to the difference in electronegativity between iodine and chlorine atoms, which produces stronger dipole–dipole forces along with London dispersion forces. On the other hand, Br_2 is a non-polar molecule and has only weak London dispersion forces between its molecules. Therefore, stronger intermolecular forces in ICl require more energy to overcome, resulting in a higher melting point compared to Br_2.

17. Define oscillatory motion.

Oscillatory Motion

Oscillatory motion is a type of periodic motion in which an object moves back and forth repeatedly about a fixed point or mean position. The motion occurs due to a restoring force that acts to bring the object back toward its equilibrium position. In oscillatory motion, the object repeats its path after a fixed interval of time. Examples include the motion of a simple pendulum, vibrations of a spring, and movement of a tuning fork. The time taken to complete one full oscillation is called the time period.

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18. What are isobaric and isochoric processes?

ISOBARIC PROCESS:

An isobaric process is a thermodynamic process in which the pressure of a system remains constant while other variables such as volume and temperature may change. In this process, heat supplied to the system can cause expansion or changes in temperature. An example is the heating of a gas in a cylinder with a movable piston under constant external pressure.

ISOCHORIC PROCESS:

An isochoric process is a thermodynamic process in which the volume of a system remains constant while pressure and temperature may change. Since the volume does not change, no work is done by or on the system. An example is heating a gas in a rigid, closed container.

19. What do you know about the Biophysical Society?

Biophysical Society

The Biophysical Society is a scientific organization that promotes the development and advancement of biophysics, a field that applies principles of physics to study biological systems. It was established in 1958 to support research and communication among scientists working in areas such as molecular biology, cellular processes, and biological structures. The society organizes scientific meetings, publishes research articles, and provides opportunities for scientists to share knowledge and collaborate. It supports research in areas such as protein structure, membrane biology, molecular interactions, and biological imaging. The Biophysical Society also encourages education and training in biophysics for students and researchers.

20. Define thermal and mechanical equilibrium.

Thermal Equilibrium

Thermal equilibrium is the condition in which two systems in contact have the same temperature and there is no net transfer of heat between them. The temperature remains constant when thermal equilibrium is reached.

Mechanical Equilibrium

Mechanical equilibrium is the condition in which a system experiences no unbalanced force or pressure difference**,** so there is no tendency for the system to move or change its mechanical state. In this condition, the pressure remains uniform throughout the system.

21. What is forced dialysis?

Forced Dialysis

Forced dialysis is a separation technique in which the movement of molecules across a semipermeable membrane is increased by applying an external force, such as pressure or an electric field. It is used to enhance the removal of small molecules, ions, and waste substances from a solution while retaining larger molecules like proteins. This technique is useful in biochemistry and medical applications, especially for the purification of biological samples and improving the efficiency of dialysis processes.

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22. Define hypotonic, isotonic and hypertonic solutions.

Hypotonic Solution

A hypotonic solution is a solution that has a lower concentration of solutes compared to the inside of a cell. When a cell is placed in a hypotonic solution, water enters the cell by osmosis, causing the cell to swell and may lead to bursting.

Isotonic Solution

An isotonic solution is a solution that has the same concentration of solutes as the cell interior. In this condition, there is no net movement of water, and the cell maintains its normal size and shape.

Hypertonic Solution

A hypertonic solution is a solution that has a higher concentration of solutes compared to the inside of a cell. When a cell is placed in a hypertonic solution, water moves out of the cell by osmosis, causing the cell to shrink.

23. Define coenzyme, cofactor, active site and prosthetic group.

Coenzyme: A coenzyme is a small organic molecule that helps an enzyme in carrying out its function. It binds with the enzyme and assists in the transfer of chemical groups or electrons during biochemical reactions. Many coenzymes are derived from vitamins.

Cofactor: A cofactor is a non-protein component required for the proper functioning of an enzyme. It may be an inorganic metal ion, such as Mg^{2+}, Zn^{2+}, or Fe^{2+}**,** or an organic molecule that helps enzyme activity.

Active Site: The active site is the specific region of an enzyme where the substrate binds and the chemical reaction takes place. It determines the specificity of the enzyme for its substrate.

Prosthetic Group: A prosthetic group is a non-protein component that is tightly and permanently attached to an enzyme and is essential for its activity. It remains bound to the enzyme during the reaction.

24. What is thermodynamics? Write its four laws.

Thermodynamics

Thermodynamics is the branch of science that deals with the study of energy, heat, work, and the relationship between them in physical and chemical systems. It explains how energy is transferred and transformed during different processes.

Laws of Thermodynamics

- ***Zeroth Law of Thermodynamics:*** It states that if two systems are separately in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.
- ***First Law of Thermodynamics:*** It states that energy cannot be created or destroyed; it can only be converted from one form to another. It represents the conservation of energy.
- ***Second Law of Thermodynamics:*** It states that the entropy (degree of disorder) of an isolated system always increases during a spontaneous process.
- ***Third Law of Thermodynamics:*** It states that the entropy of a perfectly pure crystal approaches zero as the temperature approaches absolute zero (0 K).

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25. Name groups of ion channels.

Groups of Ion Channels

Ion channels are membrane proteins that allow the movement of ions across cell membranes. They are classified into different groups based on their method of activation.

- ***Voltage-Gated Ion Channels:*** Open and close in response to changes in membrane potential. Examples include voltage-gated sodium (Na^+), potassium (K^+), and calcium (Ca^{2+}) channels.
- ***Ligand-Gated Ion Channels:*** Activated when a specific chemical molecule (ligand) binds to the channel. They are important in nerve signal transmission.
- ***Mechanically Gated Ion Channels:*** Open in response to mechanical forces such as pressure, stretch, or vibration.
- ***Leak Ion Channels:*** Remain partially open and allow ions to move across the membrane to maintain resting membrane potential.

26. Define laminar and turbulent flow dynamics.

LAMINAR FLOW

Laminar flow is a type of fluid flow in which the fluid particles move in smooth, regular, and parallel layers without mixing. It occurs at low flow rates and is characterized by a steady and orderly movement of fluid. An example is the slow flow of blood through small blood vessels.

TURBULENT FLOW

Turbulent flow is a type of fluid flow in which the fluid particles move in an irregular and chaotic manner with mixing and fluctuations in velocity. It occurs at high flow rates and is less organized than laminar flow. An example is the rapid flow of water in a river or large blood vessels.

27. What is London dispersion force?

LONDON DISPERSION FORCE

London dispersion force is a type of weak intermolecular force that occurs between all atoms and molecules due to temporary fluctuations in electron distribution. These fluctuations create temporary dipoles, which induce dipoles in nearby molecules, resulting in attractive forces between them. It is the weakest type of van der Waals force and is the only intermolecular force present in non-polar molecules. The strength of London dispersion forces increases with increasing molecular size, number of electrons, and surface area. ***Example*:** London dispersion forces exist between molecules of Br_2, I_2, and noble gases such as helium and argon.

28. Name types of molecules in the plasma membrane.

Types of Molecules in the Plasma Membrane

The plasma membrane is mainly composed of different types of molecules that help maintain the structure and function of the cell. The major types are:

- ***Phospholipids:*** Form the basic structure of the membrane as a phospholipid bilayer, providing flexibility and selective permeability.
- ***Proteins:*** Present within or attached to the membrane; they help in transport, communication, and enzymatic activities.
- ***Carbohydrates:*** Attached to proteins and lipids as glycoproteins and glycolipids; they help in cell recognition and signaling.
- ***Cholesterol:*** Maintains membrane stability, fluidity, and strength by regulating the movement of phospholipids.

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29. State types of equilibrium.

Types of Equilibrium

Equilibrium is the state of a system in which there is no net change because opposing forces or processes are balanced. The main types of equilibrium are:

- ***Thermal Equilibrium:*** Occurs when two systems have the same temperature and there is no net transfer of heat between them.
- ***Mechanical Equilibrium:*** Occurs when no unbalanced forces act on a system, and the pressure remains constant throughout the system.
- ***Chemical Equilibrium:*** Occurs when the rates of forward and reverse chemical reactions become equal, so the concentrations of reactants and products remain constant.
- ***Dynamic Equilibrium:*** A condition in which opposite processes continue at equal rates, resulting in no overall change in the system.

30. Differentiate between isotropic and anisotropic diffusion.

Difference Between Isotropic and Anisotropic Diffusion

Isotropic Diffusion	Anisotropic Diffusion
Isotropic diffusion is the movement of molecules equally in all directions.	Anisotropic diffusion is the movement of molecules that occurs more easily in one direction than others.
The rate of diffusion is the same in all directions.	The rate of diffusion varies depending on the direction.
It occurs in materials with uniform structure and properties.	It occurs in materials with different structures or properties in different directions.
Example: Diffusion of small molecules in water.	Example: Diffusion of water molecules along nerve fibers in the brain.

31. What are ball bearings? How do they reduce friction?

Ball bearings are mechanical devices consisting of small metal balls placed between two surfaces that are in relative motion. They are used to reduce friction in machines and moving parts. Ball bearings reduce friction by converting sliding friction into rolling friction, which is much smaller. The rolling motion of the balls allows the surfaces to move smoothly with less resistance and energy loss. They are commonly used in bicycle wheels, electric motors, and machine parts.

32. Define Nernst Heat Theorem.

Nernst Heat Theorem

The Nernst Heat Theorem states that as the temperature of a perfectly pure crystalline substance approaches absolute zero (0 K), the change in entropy during a reversible process approaches zero. In other words, the entropy of a perfect crystal becomes constant and reaches its minimum value at absolute zero. This theorem explains the behavior of substances at very low temperatures and provides the foundation for the Third Law of Thermodynamics. It is also used in the calculation of absolute entropy values of substances.

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33. Name solvents used in NMR.

Solvents Used in NMR

In Nuclear Magnetic Resonance (NMR) spectroscopy, special solvents called deuterated solvents are used because they contain deuterium (^2H) instead of ordinary hydrogen (^1H), which prevents interference with the NMR signal. Common solvents used in NMR are:

- ***Deuterated chloroform ($CDCl_3$):*** Most commonly used solvent for organic compounds.
- ***Deuterium oxide (D_2O):*** Used for water-soluble compounds and biological samples.
- ***Dimethyl sulfoxide-d_6 (DMSO-d_6):*** Used for polar compounds that are not soluble in $CDCl_3$.
- ***Methanol-d_4 (CD_3OD):*** Used for polar and hydrogen-bonding compounds.
- ***Acetone-d_6 ($(CD_3)_2CO$):*** Used for a variety of organic compounds.

34. Define hydrogen bonding in water.

Hydrogen Bonding in Water

Hydrogen bonding in water is an intermolecular attractive force that occurs between the hydrogen atom of one water molecule and the oxygen atom of another water molecule. It is formed because oxygen is highly electronegative and carries a partial negative charge, while hydrogen carries a partial positive charge. These hydrogen bonds give water its unique properties, such as high boiling point, surface tension, cohesion, and ability to dissolve many substances**.** Hydrogen bonding also plays an important role in maintaining the structure and function of biological molecules.

35. What is a thermometer? Write its types.

Thermometer

A thermometer is an instrument used to measure the temperature of a substance or body. It works by detecting changes in physical properties, such as expansion of liquids, electrical resistance, or radiation, that vary with temperature.

Types of Thermometers

- ***Mercury Thermometer:*** Uses the expansion of mercury in a glass tube to measure temperature.
- ***Alcohol Thermometer:*** Uses coloured alcohol and is used for measuring low temperatures.
- ***Clinical Thermometer:*** Used to measure human body temperature.
- ***Digital Thermometer:*** Measures temperature using electronic sensors and displays the result digitally.
- ***Infrared Thermometer:*** Measures temperature by detecting infrared radiation emitted from an object.

36. Explain secondary structure of protein.

Secondary Structure of Protein

The secondary structure of proteins refers to the local folding and arrangement of the polypeptide chain due to hydrogen bonding between the peptide backbone atoms. It provides stability and gives the protein chain a specific shape. The two main types of secondary structures are:

- ***Alpha (α) Helix:*** A spiral structure in which the polypeptide chain is stabilized by hydrogen bonds between amino acid residues.
- ***Beta (β) Pleated Sheet:*** A sheet-like structure formed by hydrogen bonding between different regions of the polypeptide chain.

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37. What is the difference between alpha helices and beta sheets?

Difference Between Alpha Helices and Beta Sheets

Alpha (α) Helix	Beta (β) Sheet
Alpha helix is a coiled, spiral structure of the polypeptide chain.	Beta sheet is a folded, sheet-like structure of the polypeptide chain.
It is stabilized by hydrogen bonds within the same polypeptide chain.	It is stabilized by hydrogen bonds between adjacent polypeptide strands.
The peptide backbone forms a regular right-handed spiral arrangement.	The polypeptide chains are arranged in parallel or antiparallel sheets.
It provides flexibility and compactness to proteins.	It provides strength and stability to protein structures.
Example: Keratin contains many alpha helices.	Example: Silk fibroin contains many beta sheets.

38. Name components of X-ray technique.

Components of X-ray Technique

The main components of an X-ray technique used for analysis and imaging are:

- ***X-ray Source:*** Produces X-rays using an X-ray tube or generator.
- ***Sample Holder:*** Holds and positions the sample for X-ray exposure.
- ***Monochromator/Collimator:*** Produces a focused and controlled X-ray beam.
- ***Detector:*** Detects and records the X-rays after interaction with the sample.
- ***Data Processing System:*** Analyzes the collected data & produces the final image or structural information.
- ***Display System:*** Shows the processed X-ray image or results.

39. Write applications of DNA microarray.

Applications of DNA Microarray

DNA microarray is a molecular biology technique used to study the expression and activity of thousands of genes simultaneously. It helps in analyzing genetic information and comparing gene expression patterns.

- ***Gene Expression Analysis:*** Used to measure the activity of thousands of genes under different conditions.
- ***Disease Diagnosis:*** Helps identify genetic changes associated with diseases such as cancer and inherited disorders.
- ***Drug Discovery:*** Used to study the effects of drugs on gene expression and develop new medicines.
- ***Genetic Variation Analysis:*** Helps detect mutations, genetic differences, and variations among individuals.
- ***Cancer Research:*** Used to classify different types of cancers and identify disease-related genes.
- ***Study of Biological Processes:*** Helps understand gene functions and cellular responses to environmental changes.
- ***Personalized Medicine:*** Helps in selecting suitable treatments based on an individual's genetic profile.

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40. Define system and surrounding.

System

A system is a specific part of the universe that is selected for study in thermodynamics. It may be a substance, object, or group of substances where energy and matter changes are investigated. The system is separated from the surroundings by a boundary.

Surrounding

The surroundings are everything outside the system that can interact with it by exchanging energy or matter. The interaction between the system and surroundings determines the changes that occur during a thermodynamic process.

41. How do enzymes lower activation energy?

Enzymes lower the activation energy of a biochemical reaction by providing an alternative reaction pathway that requires less energy to proceed. They bind with specific reactant molecules called substrates at their active sites and form an enzyme–substrate complex. This interaction helps to properly orient the substrates, weaken existing chemical bonds, and stabilize the transition state of the reaction. As a result, the reaction occurs more quickly and efficiently without changing the overall energy or equilibrium of the reaction. Enzymes are not consumed during the process and can be reused.

42. What are intramolecular forces of attraction?

Intramolecular Forces of Attraction

Intramolecular forces of attraction are the strong forces that exist between atoms within a molecule and hold the atoms together. These forces are responsible for the formation and stability of molecules by creating chemical bonds between atoms. The main types of intramolecular forces include:

- ***Covalent Bonds:*** Formed by the sharing of electrons between atoms.
- ***Ionic Bonds:*** Formed by the electrostatic attraction between oppositely charged ions.
- ***Metallic Bonds:*** Formed by the attraction between metal ions and free electrons in metals.

43. What are catabolic pathways?

Catabolic Pathways

Catabolic pathways are metabolic pathways in which large and complex molecules are broken down into smaller and simpler molecules. These reactions release energy, which is usually stored in the form of ATP and used by cells for various activities. During catabolism, chemical bonds of molecules such as carbohydrates, fats, and proteins are broken down through a series of enzyme-controlled reactions. Examples of catabolic pathways include cellular respiration, glycolysis, and breakdown of fatty acids.

44. Define HPLC.

HPLC

High-Performance Liquid Chromatography (HPLC) is an advanced analytical technique used for the separation, identification, purification, and quantitative analysis of components present in a mixture. In this technique, a liquid mobile phase is forced under high pressure through a column containing a stationary phase. Different compounds move through the column at different rates depending on their chemical properties and interaction with the stationary phase. HPLC is widely used in pharmaceutical industries, biochemistry, biotechnology, and clinical laboratories for the analysis of drugs, proteins, metabolites, and other biological compounds. It provides highly accurate, sensitive, and reliable results for complex mixtures.

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45. Define Entropy and Enthalpy.

Entropy

Entropy is a thermodynamic property that measures the degree of disorder or randomness of a system. It represents the distribution of energy within a system. A system with greater disorder has higher entropy, while an ordered system has lower entropy. The SI unit of entropy is joule per kelvin ($J\ K^{-1}$).

Enthalpy

Enthalpy is a thermodynamic quantity that represents the **to**tal heat content of a system at constant pressure. It is related to the internal energy of a system and the energy associated with pressure and volume. The change in enthalpy indicates the heat absorbed or released during a reaction. The SI unit of enthalpy is joule (J).

46. What is energy coupling?

Energy Coupling

Energy coupling is a process in which the energy released from one energy-producing reaction is used to drive another energy-requiring reaction. It allows cells to perform unfavorable reactions by linking them with favorable reactions that release energy. In biological systems, energy coupling commonly involves ATP **(**adenosine triphosphate**)**, which stores and transfers energy. For example, the breakdown of ATP releases energy that is used for processes such as active transport, muscle contraction, and synthesis of biomolecules**.**

47. How do bacteria undergo random walk?

Random Walk in Bacteria

Bacteria undergo random walk movement through a combination of random tumbles and directed runs. During a run, the bacterium moves in a straight direction by rotating its flagella. When it tumbles, the direction of movement changes randomly due to changes in flagellar rotation. This random movement allows bacteria to explore their environment. In the presence of chemical signals, a process called chemotaxis helps bacteria move toward favorable conditions, such as nutrients, and away from harmful substances. Thus, random walks help bacteria in searching and adapting to their surroundings.

LONG QUESTIONS

1. Explain covalent bonding.

Covalent Bonding

A covalent bond is a type of chemical bond formed when two atoms share one or more pairs of electrons to achieve a more stable electronic configuration. It usually occurs between non-metal atoms, which have similar tendencies to attract electrons and cannot easily transfer electrons from one atom to another. In a covalent bond, the shared electrons are attracted by the nuclei of both atoms, holding the atoms together. The number of shared electron pairs determines the type of covalent bond:

1. ***Single covalent bond:*** Formed when one pair of electrons is shared between two atoms.
   Example: Hydrogen molecule (H_2), where each hydrogen atom shares one electron.

2. ***Double covalent bond:*** Formed when two pairs of electrons are shared between two atoms.
   Example: Oxygen molecule (O_2), where two pairs of electrons are shared.

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3. ***Triple covalent bond:*** Formed when three pairs of electrons are shared between two atoms.
   Example: Nitrogen molecule (N_2), where three pairs of electrons are shared.

Characteristics of covalent compounds:

- They generally have low melting and boiling points compared to ionic compounds.
- They are usually poor conductors of electricity because they do not have free ions or electrons.
- Many covalent compounds exist as gases, liquids, or soft solids.
- They can form simple molecules or large structures.

2. Explain structure of RNA.

Structure of RNA

RNA (Ribonucleic Acid) is a nucleic acid that plays important roles in protein synthesis, gene regulation, and the transfer of genetic information**.** It is usually single-stranded and is made up of repeating units called nucleotides. Each RNA nucleotide consists of three components:

- ***Ribose sugar:*** RNA contains a five-carbon sugar called ribose. The presence of ribose distinguishes RNA from DNA, which contains deoxyribose sugar.
- ***Phosphate group:*** The phosphate group links nucleotides together through phosphodiester bonds, forming the backbone of the RNA strand.
- ***Nitrogenous bases:*** RNA contains four nitrogenous bases:
    - Adenine (A)
    - Guanine (G)
    - Cytosine (C)
    - Uracil (U**)** (replaces thymine found in DNA)

The RNA strand has a sugar-phosphate backbone with nitrogenous bases attached to the sugar. Although RNA is generally single-stranded, it can fold into complex shapes due to base pairing between complementary bases:

- Adenine pairs with Uracil (A–U)
- Guanine pairs with Cytosine (G–C)

3. Explain Newton's third law of motion.

Newton's Third Law of Motion

Newton's Third Law of Motion states that:

"For every action, there is an equal and opposite reaction."

This means that whenever one object applies a force on another object, the second object applies an equal force in the opposite direction on the first object. These two forces always occur together and act on different objects.

Examples of Newton's Third Law:

- ***Walking:*** When a person walks, their feet push the ground backward (action). The ground pushes the person forward with an equal and opposite force (reaction), allowing them to move.
- ***Rocket motion:*** A rocket pushes gases downward with great force (action). The gases push the rocket upward with an equal and opposite force (reaction), causing the rocket to rise.

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Importance of Newton's Third Law:

- It explains how forces work in everyday activities.
- It helps explain the movement of vehicles, rockets, and living organisms.
- It shows that forces always occur in pairs.

4. Write derivation of kinetic friction.

Derivation of Kinetic Friction

Kinetic friction is the force that opposes the motion of an object when it slides over a surface. It depends on the nature of the surfaces in contact and the normal force acting between them. The force of kinetic friction is given by:

$$F_k = u_k\, N$$

Where:

- F_k = Kinetic frictional force
- u_k = Coefficient of kinetic friction
- N = Normal reaction force

Derivation:

Consider a body of mass 'm' sliding on a horizontal surface.

1. The weight of the body acts downward:

$$W = mg$$

Where '**g**' is the acceleration due to gravity.

2. The surface exerts an equal and opposite normal reaction force:

$$N = mg$$

3. According to experiments, the kinetic frictional force is directly proportional to the normal reaction:

$$F_k \propto N$$

Removing the proportionality sign by introducing a constant:

$$F_k = u_k\, N$$

Where 'u_k' is the coefficient of kinetic friction.

4. Substituting the value of normal force:

$$F_k = u_k\, mg$$

Therefore, the kinetic frictional force acting on a sliding object is:

$$F_k = u_k N$$

5. Explain Dissociation Constant of Water.

Dissociation Constant of Water (Ion Product of Water, (K_w)

The dissociation constant of water is the equilibrium constant for the self-ionization of water. In this process, two water molecules react to form hydrogen ions and hydroxide ions. The reaction is:

$$2H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + OH^-_{(aq)}$$

The ion product (dissociation constant) of water is given by:

$$K_w = [H^+][OH^-]$$

At 25°C (298 K)**:**

- $[H^+]$ = 1.0 x 10^{-7} mol/L
- $[OH^-]$ = 1.0 x 10^{-7} mol/L

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Therefore,

$$K_w = (1.0\ x\ 10^{-7})(1.0\ x\ 10^{-7}) = 1.0\ x\ 10^{-14}$$

This constant is used to determine whether a solution is acidic, basic, or neutral:

- ***Neutral:*** $[H^+] = [OH^-]$
- ***Acidic:*** $[H^+] > [OH^-]$
- ***Basic (Alkaline):*** $[H^+] < [OH^-]$

6. Write a detailed note on Imaging.

- ***Imaging:*** Imaging is the process of creating visual representations of objects, structures, or body parts using different techniques. In medicine, imaging is used to examine internal organs, detect diseases, and help in diagnosis and treatment planning.
- ***X-ray Imaging:*** X-ray imaging uses electromagnetic radiation to produce pictures of internal body structures. It is mainly used to detect bone fractures, dental problems, and certain abnormalities. Dense structures like bones absorb more X-rays and appear white in the image.
- ***Ultrasound Imaging:*** Ultrasound imaging uses high-frequency sound waves to create images of internal organs and tissues. It is commonly used to monitor pregnancy, examine the heart, and evaluate organs such as the liver and kidneys. It is a safe and non-invasive technique.
- ***Computed Tomography (CT) Imaging:*** CT imaging combines multiple X-ray images taken from different angles to produce detailed cross-sectional images of the body. It is used to identify tumors, internal injuries, and disorders of organs such as the brain and lungs.
- ***Magnetic Resonance Imaging (MRI):*** MRI uses strong magnetic fields and radio waves to create detailed images of soft tissues in the body. It is useful for examining the brain, spinal cord, muscles, and joints, and it does not use ionizing radiation.
- ***Nuclear Imaging:*** Nuclear imaging uses small amounts of radioactive substances called tracers to show the function of organs and tissues. It is used in techniques such as PET and SPECT scans to diagnose diseases related to the heart, bones, and other organs.

7. Explain dipole–dipole forces in ICl (Iodine Chloride) molecules.

Dipole–Dipole Forces in ICl (Iodine Chloride) Molecules

Iodine chloride (ICl) is a polar molecule because iodine and chlorine atoms have different electronegativities. Chlorine is more electronegative than iodine, so the chlorine atom develops a partial negative charge and the iodine atom develops a partial positive charge. This creates a permanent dipole in the ICl molecule. Dipole–dipole forces occur due to the attraction between the positive end of one ICl molecule and the negative end of another ICl molecule. The iodine end of one molecule is attracted to the chlorine end of a neighboring molecule. These intermolecular forces help hold ICl molecules together and influence its physical properties, such as boiling point and melting point. In ICl, dipole–dipole forces exist along with weaker London dispersion forces. Because ICl is a polar molecule, dipole–dipole interactions make a significant contribution to its intermolecular attraction.

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8. Explain basic components of the cell membrane.

Basic Components of the Cell Membrane

The cell membrane, also known as the plasma membrane, is a thin, flexible, selectively permeable membrane that surrounds the cell and separates the internal environment of the cell from the external environment. It controls the movement of substances in and out of the cell and plays an important role in maintaining cell structure and communication. According to the fluid mosaic model, the cell membrane is composed of a mixture of lipids, proteins, carbohydrates, and cholesterol arranged in a flexible structure.

- ***Phospholipids:*** Phospholipids are the major components of the cell membrane and form a double-layered structure called the phospholipid bilayer. Each phospholipid molecule has a hydrophilic (water-loving) head and hydrophobic (water-repelling) tails. The hydrophilic heads face the watery environments inside and outside the cell, while the hydrophobic tails remain inside the membrane. This arrangement forms a barrier that controls the entry and exit of substances.
- ***Membrane Proteins:*** Proteins are present within the phospholipid bilayer or attached to its surface. These proteins perform several important functions, including transport of ions and molecules, acting as receptors for signals, functioning as enzymes, and providing structural support. Transport proteins help substances move across the membrane through channels and carriers.
- ***Cholesterol:*** Cholesterol molecules are found between phospholipids in the cell membrane. They help maintain the stability and fluidity of the membrane. Cholesterol prevents the membrane from becoming too stiff in cold conditions and too fluid in warm conditions, allowing the cell membrane to function properly.
- ***Carbohydrates:*** Carbohydrates are found on the outer surface of the cell membrane attached to proteins and lipids, forming glycoproteins and glycolipids. They are involved in cell recognition, communication, and identification of cells by the immune system.
- ***Glycoproteins and Glycolipids:*** Glycoproteins are proteins with carbohydrate chains attached, while glycolipids are lipids with carbohydrate chains. They help cells recognize each other, attach to surrounding structures, and communicate with neighboring cells.

9. Define exergonic and endergonic reactions. Also define free energy.

Free Energy

Free energy is the amount of energy available in a system that can be used to perform work. It is the energy present in molecules that can be released or used during a chemical reaction. In biological systems, free energy determines whether a reaction can occur spontaneously or requires an input of energy. It is represented by the symbol **G** and the change in free energy is represented as ΔG.

Exergonic Reactions

Exergonic reactions are chemical reactions that release free energy to the surroundings. These reactions occur spontaneously because the products contain less free energy than the reactants. In exergonic reactions, the value of ΔG is negative ($-\Delta G$)**.** An example is the breakdown of glucose during cellular respiration, where energy is released and used by cells.

Endergonic Reactions

Endergonic reactions are chemical reactions that require an input of free energy from the surroundings to occur. These reactions are non-spontaneous because the products have more free energy than the reactants. In endergonic reactions, the value of ΔG is positive ($+\Delta G$)**.** An example is photosynthesis, where plants use light energy to produce glucose from carbon dioxide and water.

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10. Name and explain five intensive properties.

Five Intensive Properties

Intensive properties are physical properties of a substance that do not depend on the amount of substance present. These properties remain the same whether the sample size is large or small.

- ***Temperature:*** Temperature is a measure of the average kinetic energy of particles in a substance. It indicates how hot or cold a substance is and does not change with the amount of material present.
- ***Density:*** Density is the mass of a substance per unit volume. It is calculated by dividing mass by volume. The density of a pure substance remains constant regardless of the quantity of the substance.
- ***Pressure:*** Pressure is the force exerted by a substance per unit area. It is an intensive property because it remains the same even if the amount of substance is increased or decreased under the same conditions.
- ***Melting Point:*** Melting point is the temperature at which a solid changes into a liquid. It is a characteristic property of a substance and does not depend on the amount of sample.
- ***Boiling Point:*** Boiling point is the temperature at which a liquid changes into a gas. It is an intensive property because it remains constant for a pure substance at a given pressure.

11. Write the principle of centrifugation.

Principle of Centrifugation

Centrifugation is a laboratory technique used for the separation of particles suspended in a fluid by applying a strong centrifugal force. It is based on the principle that particles of different sizes, shapes, and densities sediment at different rates when subjected to centrifugal force.

When a mixture is placed in a centrifuge and rotated at high speed, a centrifugal force acts on the particles. The heavier and denser particles move outward and settle at the bottom of the centrifuge tube, forming a pellet, while the lighter particles remain in the liquid portion called the supernatant. The rate of sedimentation depends on the size, mass, and density of particles, as well as the speed and duration of centrifugation.

Centrifugation is widely used in biological and chemical laboratories for the separation and purification of cells, cell organelles, proteins, nucleic acids, blood components, and other biological molecules. Different types of centrifugation, such as differential centrifugation and density gradient centrifugation**,** are used depending on the nature of the material being separated.

12. How does diffusion occur in lungs and intestine?

Diffusion is a passive transport process in which molecules move from a region of higher concentration to a region of lower concentration across a selectively permeable membrane. This movement occurs due to the random motion of molecules and does not require cellular energy (ATP). In biological systems, diffusion plays an essential role in the exchange of gases in the lungs and absorption of nutrients in the intestine.

Diffusion in the Lungs

In the lungs, diffusion occurs across the alveolar-capillary membrane, which separates the air inside the alveoli from the blood in surrounding pulmonary capillaries. The exchange of respiratory gases depends on differences in their partial pressures.

1. Oxygen diffusion: Oxygen concentration is higher in the alveolar air than in the deoxygenated blood arriving through pulmonary arteries. Due to this concentration gradient, oxygen diffuses from the alveoli into the blood through the thin alveolar membrane. It then binds with hemoglobin inside red blood cells and is transported to body tissues.

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2. Carbon dioxide diffusion: Carbon dioxide concentration is higher in the blood than in the alveoli. Therefore, carbon dioxide diffuses from the blood into the alveoli and is removed from the body during exhalation. The efficiency of gas diffusion in the lungs is increased by:

- Large surface area of alveoli
- Very thin alveolar walls
- Rich blood supply through pulmonary capillaries
- Continuous maintenance of concentration gradients through breathing and circulation

Diffusion in the Intestine

In the intestine, diffusion is involved in the movement of digested nutrients and other substances from the intestinal lumen into the bloodstream. The inner surface of the small intestine contains numerous villi and microvilli, which greatly increase the surface area available for absorption.

1. Diffusion of nutrients: Small, lipid-soluble molecules such as fatty acids, glycerol, and some vitamins can pass directly through the intestinal epithelial cells by simple diffusion. They move from an area of higher concentration in the intestinal lumen to an area of lower concentration in intestinal cells or blood.

2. Facilitated diffusion: Some nutrients, such as certain sugars and amino acids, require transport proteins present in the intestinal membrane. These molecules move down their concentration gradient through carrier proteins without the use of ATP.

3. Factors affecting intestinal diffusion: Concentration gradient between intestinal contents and blood

- Surface area provided by villi and microvilli
- Thickness and permeability of the intestinal membrane
- Availability of transport proteins

13. Differentiate between shielding and deshielding of protons.

Difference Between Shielding and Deshielding of Protons

Shielding	Deshielding
Shielding occurs when electrons surrounding a proton create an induced magnetic field that opposes the external magnetic field applied during NMR analysis.	Deshielding occurs when electron density around a proton decreases, causing the proton to experience a stronger external magnetic field.
A shielded proton is protected from the external magnetic field due to higher electron density around it.	A deshielded proton experiences a greater effect of the external magnetic field because of reduced electron density.
Shielded protons absorb energy at a lower frequency and appear upfield (towards lower chemical shift values) in an NMR spectrum.	Deshielded protons absorb energy at a higher frequency and appear downfield (towards higher chemical shift values) in an NMR spectrum.
Electron-donating groups increase electron density and cause proton shielding.	Electronegative atoms or electron-withdrawing groups decrease electron density and cause proton deshielding.
Example: Protons in simple alkyl groups such as **CH_3–** are relatively shielded due to higher electron density.	Example: Protons near electronegative atoms such as oxygen in **CH_3–O–H** are deshielded due to electron withdrawal by oxygen.

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14. Explain the principle of Mass Spectrometer.

Principle of Mass Spectrometer

A mass spectrometer is an analytical instrument used to determine the mass-to-charge ratio (m/z) of ions and is used for the identification and analysis of chemical compounds. It works on the principle that charged particles (ions) can be separated according to their mass-to-charge ratio when they are subjected to electric and magnetic fields**.**

In a mass spectrometer, the sample is first converted into gaseous ions by a process called ionization. The produced ions are accelerated by an electric field, giving them kinetic energy. These fast-moving ions then enter a magnetic field, where they experience a force that causes them to follow curved paths. The extent of deflection depends on the mass-to-charge ratio of the ions; lighter ions or ions with greater charge are deflected more than heavier ions or ions with lower charge.

The separated ions are detected by an ion detector, which records their abundance and generates a mass spectrum. The mass spectrum contains peaks corresponding to different ions, where the position of each peak represents the m/z value and the intensity represents the relative abundance of that ion. The basic principle can be summarized as:

Ionization → Acceleration of ions → Separation according to m/z ratio → Detection and analysis of ions.

Mass spectrometry is widely used for determining molecular masses, identifying unknown compounds, studying molecular structures, and analyzing biological molecules such as proteins, peptides, and nucleic acids.

15. Write types of ultra-centrifugation.

Types of Ultracentrifugation

Ultracentrifugation is an advanced centrifugation technique that uses extremely high centrifugal forces to separate very small particles such as viruses, ribosomes, proteins, nucleic acids, and cell organelles based on their size, shape, and density. There are two major types of ultracentrifugation:

1. Differential Ultracentrifugation

Differential ultracentrifugation is a method in which a cell or tissue homogenate is subjected to a series of centrifugation steps at progressively increasing speeds. During each step, particles with larger size and higher density sediment first, forming a pellet, while smaller and less dense particles remain in the supernatant.

In this technique, different cellular components are separated sequentially. For example, nuclei and large cell debris sediment at lower speeds, while mitochondria, lysosomes, and ribosomes require higher speeds for separation. It is commonly used for isolation of cell organelles.

2. Density Gradient Ultracentrifugation

Density gradient ultracentrifugation separates particles according to their buoyant density by using a density gradient medium such as sucrose or cesium chloride. During centrifugation, particles move through the gradient until they reach a position where their density is equal to the surrounding medium.
This method provides highly purified separation of particles with similar sizes but different densities. It is widely used for the purification of viruses, DNA, RNA, proteins, and subcellular components.

Types of Density Gradient Ultracentrifugation:

a) Rate-Zonal Centrifugation: In this method, particles are separated mainly according to their size and mass. The sample is placed on top of a preformed density gradient, and particles move at different rates during centrifugation. Separation depends on sedimentation velocity.

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b) Isopycnic Centrifugation: In this method, particles are separated according to their density. Each particle migrates to a position in the density gradient where its density equals that of the surrounding medium. This technique is commonly used for separation of nucleic acids and viruses.

16. Explain classification of amino acids.

Classification of Amino Acids

Amino acids are organic compounds containing an amino group ($–NH_2$) and a carboxyl group (–COOH) attached to the same carbon atom (α-carbon). They are the basic structural units of proteins. Amino acids are classified in several ways based on their chemical properties, nutritional importance, and structure.

1. Classification Based on Side Chain (R-group) Properties

a) Non-polar (Hydrophobic) Amino Acids: These amino acids contain non-polar side chains and are generally insoluble in water. They are usually located inside the protein structure, away from the aqueous environment. ***Examples*:** Glycine, Alanine, Valine, Leucine, Isoleucine, Methionine, Proline, Phenylalanine, Tryptophan.

b) Polar Uncharged Amino Acids: These amino acids contain polar side chains that can form hydrogen bonds but do not carry a net electrical charge at physiological pH. ***Examples*:** Serine, Threonine, Cysteine, Tyrosine, Asparagine, Glutamine.

c) Acidic Amino Acids: These amino acids contain additional carboxyl groups in their side chains and carry a negative charge at physiological pH. ***Examples*:** Aspartic acid and Glutamic acid.

d) Basic Amino Acids: These amino acids contain additional amino groups or nitrogen-containing groups in their side chains and carry a positive charge at physiological pH. ***Examples*:** Lysine, Arginine, and Histidine.

2. Classification Based on Nutritional Requirement

a) Essential Amino Acids: Essential amino acids cannot be synthesized by the human body in sufficient amounts and must be obtained through diet. ***Examples*:** Valine, Leucine, Isoleucine, Lysine, Methionine, Threonine, Tryptophan, Phenylalanine, and Histidine.

b) Non-Essential Amino Acids: Non-essential amino acids can be synthesized by the body through metabolic pathways and do not need to be supplied in the diet. ***Examples*:** Alanine, Aspartic acid, Glutamic acid, Serine, and Glycine.

c) Conditionally Essential Amino Acids: These amino acids are normally synthesized by the body but become essential during periods of growth, illness, stress, or metabolic disorders. ***Examples*:** Arginine, Cysteine, Glutamine, Tyrosine, and Proline.

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3. Classification Based on Metabolic Fate

a) Glucogenic Amino Acids: These amino acids are converted into glucose or glucose-producing intermediates through gluconeogenesis. ***Examples*:** Alanine, Glycine, Aspartate, and Glutamate.

b) Ketogenic Amino Acids: These amino acids are converted into ketone bodies or acetyl-CoA. ***Examples*:** Leucine and Lysine.

c) Both Glucogenic and Ketogenic Amino Acids: Some amino acids can produce both glucose and ketone bodies. ***Examples*:** Phenylalanine, Tyrosine, Tryptophan, Isoleucine, and Threonine.

4. Classification Based on Number of Amino and Carboxyl Groups

a) Neutral Amino Acids: These contain one amino group and one carboxyl group. ***Examples*:** Glycine, Alanine, Valine.

b) Basic Amino Acids: These contain more than one amino group. ***Examples*:** Lysine, Arginine, Histidine.

c) Acidic Amino Acids: These contain more than one carboxyl group. ***Examples*:** Aspartic acid and Glutamic acid.

17. Write uses of ultra-filtration.

Uses of Ultrafiltration

Ultrafiltration is a membrane-based separation technique in which molecules are separated according to their size and molecular weight using a semi-permeable membrane. A pressure gradient is applied to force smaller molecules and solvents through the membrane, while larger molecules are retained.

- ***Purification and Concentration of Proteins:*** Ultrafiltration is widely used in biochemistry and biotechnology for the separation, purification, and concentration of proteins, enzymes, and other macromolecules. The membrane retains large protein molecules while allowing small molecules, salts, and solvents to pass through.
- ***Separation of Biological Molecules:*** It is used to separate and purify biological substances such as DNA, RNA, antibodies, polysaccharides, and other macromolecules based on their molecular size.
- ***Water and Wastewater Treatment:*** Ultrafiltration is used for purification of drinking water and treatment of industrial wastewater. It removes suspended particles, bacteria, viruses, colloids, and other impurities while allowing water and small dissolved substances to pass through.
- ***Pharmaceutical Applications:*** In pharmaceutical industries, ultrafiltration is used for the purification and concentration of drug molecules, vaccine components, and biological products. It helps maintain product quality and sterility.
- ***Food and Dairy Industry:*** Ultrafiltration is used in the processing of milk and other food products. It helps concentrate milk proteins, produce whey protein concentrates, and remove unwanted components while retaining valuable nutrients.
- ***Removal of Pyrogens and Microorganisms:*** Ultrafiltration membranes can remove bacteria, endotoxins, and other contaminants from biological solutions and laboratory preparations.
- ***Laboratory Sample Preparation:*** It is used in research laboratories to concentrate dilute samples, exchange buffers, and prepare biological samples for further analysis.

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18. Explain Quasi Equilibrium.

Quasi-Equilibrium

Quasi-equilibrium is a state in which a system is not in complete equilibrium but remains very close to equilibrium because the changes occurring within the system are extremely slow. In this condition, the system passes through a series of intermediate equilibrium states, allowing equilibrium equations to be applied approximately at each stage. In thermodynamics, a quasi-equilibrium process occurs when the system changes so slowly that the internal properties such as pressure, temperature, and volume remain uniform throughout the system at any given moment. The system has enough time to adjust to the changes, and each intermediate state can be considered an equilibrium state. For a process to be considered quasi-equilibrium:

- The change must occur gradually and very slowly.
- The system must remain nearly uniform at every stage.
- The difference between the system and surroundings must be very small.
- Internal equilibrium must be maintained during the process.

A common example of a quasi-equilibrium process is the slow compression or expansion of a gas in a piston-cylinder system. If the piston moves very slowly, the gas pressure and temperature adjust continuously, and the gas remains almost at equilibrium throughout the process. Quasi-equilibrium is an idealized concept used in thermodynamics because true equilibrium requires no change, while real processes always involve some degree of imbalance. It allows scientists and engineers to analyze complex processes using equilibrium relationships with reasonable accuracy.

19. Differentiate between macroassembly and microassembly.

Difference Between Macroassembly and Microassembly

Macroassembly	Microassembly
Macroassembly is the process of assembling large-scale structures or components that are visible to the naked eye.	Microassembly is the process of assembling very small components at the micrometer or nanometer scale.
It involves the joining or arrangement of macroscopic parts such as mechanical components, devices, or biological structures.	It involves the precise manipulation and placement of microscopic components such as microchips, cells, molecules, or microelectromechanical systems (MEMS).
Assembly is usually performed using conventional tools and human-operated techniques.	Assembly requires specialized techniques such as microscopy, robotics, micro-manipulation, and advanced fabrication methods.
The accuracy and precision required are relatively lower because the components are larger.	It requires extremely high accuracy and precision due to the small size of the components.
Examples include assembling machines, electronic devices, and mechanical systems.	Examples include fabrication of microelectronic circuits, assembly of MEMS devices, and arrangement of biological molecules or nanoparticles.
The forces involved are mainly mechanical forces such as gravity, friction, and mechanical bonding.	The forces involved include surface forces, electrostatic interactions, van der Waals forces, and molecular interactions.

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20. State and explain heat transfer.

Heat Transfer

Heat transfer is the process by which thermal energy is transferred from a region of higher temperature to a region of lower temperature due to a temperature difference. Heat transfer continues until thermal equilibrium is reached between the interacting systems. The rate and mechanism of heat transfer depend on the properties of the materials involved and the temperature gradient. There are three main modes of heat transfer:

1. Conduction

Conduction is the transfer of heat through a substance without the actual movement of the particles of the substance. Heat is transferred from the hotter region to the colder region through molecular vibrations and collisions between adjacent particles. In solids, especially metals, conduction occurs mainly due to the movement of free electrons and molecular vibrations. Materials such as metals are good conductors of heat, while materials like wood and plastic are poor conductors. ***Example*:** Heat transfer through a metal rod when one end is heated.

2. Convection

Convection is the transfer of heat through the movement of fluids (liquids and gases). When a fluid is heated, its density decreases, causing it to rise, while cooler and denser fluid moves downward. This continuous movement creates convection currents that transfer heat. Convection is of two types:

- ***Natural convection:*** Occurs due to density differences caused by temperature variations.
- ***Forced convection:*** Occurs when fluid movement is produced by external devices such as pumps or fans.

***Example*:** Circulation of water during heating and movement of warm air in a room.

3. Radiation

Radiation is the transfer of heat energy in the form of electromagnetic waves, mainly infrared radiation. It does not require a medium and can occur through empty space. All objects emit thermal radiation, and the amount of heat transferred depends on the temperature and surface properties of the object. ***Example*:** Heat received from the Sun reaches the Earth through radiation.

21. Explain instrumentation of NMR.

Instrumentation of NMR (Nuclear Magnetic Resonance) Spectrometer

NMR instrumentation refers to the arrangement of components used in a Nuclear Magnetic Resonance spectrometer to produce, detect, and analyze the signals generated by atomic nuclei placed in a magnetic field. The instrument works on the principle that nuclei with magnetic properties absorb and re-emit electromagnetic radiation when placed in a strong magnetic field. The major components of an NMR spectrometer are:

1. Strong Magnetic Field and Magnet System

The magnet is the most important component of an NMR spectrometer. It produces a strong and uniform magnetic field in which the sample is placed. The magnetic field aligns the nuclear spins of atoms such as ^{1}H, ^{13}C, and ^{31}P in different energy states. Modern NMR instruments commonly use superconducting magnets that provide highly stable and powerful magnetic fields.

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2. Sample Holder and Probe

The sample is placed in a small glass tube called an NMR tube, which is positioned inside the probe. The probe contains the radiofrequency (RF) coil that transmits electromagnetic radiation to the sample and detects the resulting NMR signals. The probe also maintains the correct position and temperature of the sample during analysis.

3. Radiofrequency Transmitter

The radiofrequency transmitter generates electromagnetic pulses at a specific frequency that are applied to the nuclei in the magnetic field. These RF pulses provide energy to excite the nuclei, causing them to move from a lower energy state to a higher energy state.

4. Radiofrequency Receiver and Detector

After excitation, nuclei return to their original energy state and release energy as radiofrequency signals. The receiver detects these weak signals and converts them into electrical signals. These signals contain information about the chemical environment of the nuclei.

5. Amplifier

The detected NMR signals are very weak, so an amplifier increases their strength before further processing. This improves the sensitivity and quality of the NMR spectrum.

6. Computer System and Data Processor

The computer system processes the electrical signals obtained from the detector and converts them into an NMR spectrum. It performs functions such as signal amplification, Fourier transformation, peak analysis, and calculation of chemical shifts.

7. Shim System

The shim system consists of small correction coils that improve the uniformity of the magnetic field. A highly homogeneous magnetic field is necessary to obtain sharp and accurate NMR signals.

8. Lock System

The lock system maintains the stability of the magnetic field during the experiment. It usually uses a reference signal from a solvent nucleus, such as deuterium (^2H), to correct small changes in the magnetic field.

22. What is Brownian motion? Give two reasons why Albert Einstein mentioned it in 1905.

Brownian Motion

Brownian motion is the continuous, random, and irregular movement of microscopic particles suspended in a fluid (liquid or gas) due to their constant collision with the rapidly moving molecules of the surrounding medium. The motion appears as a zigzag pattern because the suspended particles are continuously pushed in different directions by the invisible molecules of the fluid.

The phenomenon was first observed by the botanist Robert Brown in 1827 while studying pollen grains suspended in water. Later, Albert Einstein explained the mathematical basis of Brownian motion in 1905 by relating it to the molecular theory of matter.

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Reasons Why Albert Einstein Mentioned Brownian Motion in 1905

1. Evidence for the Existence of Atoms and Molecules:

Einstein used Brownian motion as strong evidence that atoms and molecules actually exist. The random movement of suspended particles could be explained only by assuming that they were being continuously bombarded by invisible molecules of the fluid.

2. Confirmation of Molecular Theory and Calculation of Molecular Properties:

Einstein developed a mathematical theory of Brownian motion that allowed scientists to calculate molecular dimensions, the number of molecules present in a given amount of substance (Avogadro's number), and verify the kinetic theory of matter. His work provided experimental support for the idea that matter is composed of constantly moving particles.

23. Explain endothermic and exothermic reactions.

Chemical reactions involve the conversion of energy, mainly in the form of heat, between the reacting system and its surroundings. Based on the exchange of heat energy, chemical reactions are classified into endothermic and exothermic reactions.

1. Endothermic Reactions

An endothermic reaction is a chemical reaction in which heat energy is absorbed from the surroundings by the reacting system. In these reactions, the energy required to break the bonds of reactant molecules is greater than the energy released during the formation of product molecules. As a result, the system gains energy and the surroundings become cooler. In an endothermic reaction, the enthalpy change **(**ΔH**)** is positive because heat is absorbed:

$$\Delta H > 0$$

Characteristics of Endothermic Reactions:

- Heat is absorbed from the surroundings.
- Products contain more energy than reactants.
- The temperature of the surroundings decreases.
- Energy is required for the reaction to proceed.
- The reaction has a positive enthalpy change.

Examples:

1. ***Photosynthesis:***

$$6CO_2 + 6H_2O + Light\ Energy \rightarrow C_6H_{12}O_6 + 6H_2O$$

Plants absorb light energy to convert carbon dioxide and water into glucose.

2. ***Thermal decomposition of calcium carbonate:***

$$CaCO_3 + heat \rightarrow CaO + CO_2$$

Heat energy is required to break down calcium carbonate.

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2. Exothermic Reactions

An exothermic reaction is a chemical reaction in which heat energy is released from the reacting system to the surroundings. In these reactions, the energy released during the formation of new bonds in products is greater than the energy required to break the bonds in reactants. Therefore, excess energy is released as heat. In an exothermic reaction, the enthalpy change **(**ΔH**)** is negative because heat is released:

$$\Delta H < 0$$

Characteristics of Exothermic Reactions:

- Heat is released into the surroundings.
- Products contain less energy than reactants.
- The temperature of the surroundings increases.
- Energy is produced during the reaction.
- The reaction has a negative enthalpy change.

Examples:

1. ***Combustion of methane:***

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + heat$$

Methane burns and releases heat energy.

2. ***Neutralization reaction:***

$$HCl + NaOH \rightarrow NaCl + H_2O + heat$$

Heat is released when an acid reacts with a base.

24. Explain all four laws of thermodynamics.

Laws of Thermodynamics

Thermodynamics is the branch of physical science that deals with the relationships between heat, energy, work, and the transformations of energy in a system. The laws of thermodynamics describe the fundamental principles governing energy transfer and energy changes.

1. Zeroth Law of Thermodynamics

The Zeroth Law of Thermodynamics states that if two systems are individually in thermal equilibrium with a third system, then they are also in thermal equilibrium with each other. This law establishes the concept of temperature and defines thermal equilibrium. When two systems have the same temperature, there is no net transfer of heat between them. Mathematically,

$$T_A = T_C \quad and \quad T_B = T_C \implies T_A = T_B$$

2. First Law of Thermodynamics

The First Law of Thermodynamics states that energy can neither be created nor destroyed; it can only be transformed from one form to another**.** It is also known as the law of conservation of energy. According to this law, the heat supplied to a system is used to increase the internal energy of the system and/or perform work on the surroundings. Mathematically,

$$\Delta U = Q - W$$

Where:

- ΔU = change in internal energy
- Q = heat supplied to the system
- W = work done by the system

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3. Second Law of Thermodynamics

The Second Law of Thermodynamics states that heat transfer occurs naturally from a region of higher temperature to a region of lower temperature, and no spontaneous process can reverse this direction without external work**.** This law introduces the concept of entropy, which represents the degree of randomness or disorder in a system. For a spontaneous process, the total entropy of an isolated system increases. Mathematically,

$$\Delta S_{universe} > 0$$

Where **ΔS** represents the change in entropy.

4. Third Law of Thermodynamics

The Third Law of Thermodynamics states that the entropy of a perfectly crystalline substance approaches zero as the temperature approaches absolute zero (0 K)**.** At absolute zero, molecular motion becomes minimum and the system reaches its lowest possible energy state. Mathematically,

$$T \rightarrow 0K, \quad S \rightarrow 0$$

Where:

- T = temperature
- S = entropy

25. Differentiate between renewable and non-renewable sources of energy.

Difference Between Renewable and Non-Renewable Sources of Energy

Renewable Sources of Energy	Non-Renewable Sources of Energy
Renewable energy sources are those sources of energy that are naturally replenished and can be used repeatedly without being exhausted.	Non-renewable energy sources are those sources of energy that exist in limited amounts and cannot be replaced within a short period after use.
They are obtained from natural processes that are continuously available.	They are obtained from finite natural resources formed over millions of years.
They have a lower environmental impact and generally produce less pollution and greenhouse gases.	They usually produce higher levels of pollution and greenhouse gas emissions during extraction and use.
Their availability is generally sustainable for long-term energy production.	Their availability decreases continuously with consumption and may eventually become depleted.
Energy production from renewable sources depends on natural conditions such as sunlight, wind, and water availability.	Energy production from non-renewable sources depends on the extraction and processing of limited fuel reserves.
They require advanced technology and infrastructure for efficient energy conversion.	They are widely used because of established technology and high energy density.
Examples include solar energy, wind energy, hydroelectric energy, geothermal energy, and biomass energy.	Examples include coal, petroleum, natural gas, and nuclear fuels such as uranium.

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26. What is emission spectroscopy?

Emission Spectroscopy

Emission spectroscopy is an analytical technique used to study the electromagnetic radiation emitted by atoms, ions, or molecules when they return from an excited energy state to a lower energy state. The emitted radiation is analyzed according to its wavelength or frequency to identify and measure the components present in a sample.

When atoms or molecules absorb energy from an external source, their electrons become excited and move to higher energy levels. Since the excited state is unstable, electrons return to lower energy levels and release the absorbed energy in the form of photons. The wavelength of emitted light corresponds to the difference between the energy levels of the electrons.

The emitted radiation is passed through a spectrometer, which separates it into individual wavelengths to produce an emission spectrum. Each element produces a characteristic spectrum because each atom has unique energy levels. The position and intensity of spectral lines provide information about the identity and concentration of elements present in the sample.

Emission spectroscopy is based on the principle that excited atoms emit radiation of specific wavelengths when electrons return to lower energy states**.** It is widely used for elemental analysis and identification of chemical substances.

27. What is ionic bonding?

Ionic Bonding

Ionic bonding is a type of chemical bond formed by the complete transfer of one or more electrons from one atom to another, resulting in the formation of oppositely charged ions that are held together by strong electrostatic forces of attraction. It is also known as an electrovalent bond. Ionic bonding generally occurs between metals and non-metals. Metals have a tendency to lose electrons due to their lower ionization energy, forming positively charged ions (cations). Non-metals have a tendency to gain electrons because of their higher electron affinity and electronegativity, forming negatively charged ions (anions). The attraction between these oppositely charged ions leads to the formation of an ionic compound.

Formation of Ionic Bond

During ionic bond formation, one atom donates electrons while another atom accepts those electrons to achieve a stable electronic configuration. For example, in sodium chloride (NaCl):

$$Na \rightarrow Na^{+} + e^{-}$$

Sodium loses one electron to form a positively charged sodium ion.

$$Cl + e^{-} \rightarrow Cl^{-}$$

Chlorine gains one electron to form a negatively charged chloride ion.

The resulting ions are held together by electrostatic attraction:

$$Na^{+} + Cl^{-} \rightarrow NaCl$$

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Nature of Ionic Bond

The strength of an ionic bond depends on the magnitude of electrostatic attraction between the ions. It increases with:

- Higher charges on the ions.
- Smaller distance between the oppositely charged ions.
- Smaller ionic size.

Ionic compounds usually form a three-dimensional crystal lattice structure in which each cation is surrounded by anions and each anion is surrounded by cations.

Properties of Ionic Compounds

- Ionic compounds generally have high melting and boiling points due to strong electrostatic forces.
- They are usually crystalline solids at room temperature.
- They conduct electricity when dissolved in water or in molten form because ions become free to move.
- They are generally soluble in polar solvents such as water.
- They are hard but often brittle due to the arrangement of ions in the crystal lattice.

28. What are hormones?

Hormones

Hormones are chemical messengers produced and secreted by specialized cells or endocrine glands that regulate and coordinate various physiological processes in the body. They are released into the bloodstream and transported to specific target cells or organs, where they produce a response by binding to specific receptors. Hormones are effective in very small concentrations and play important roles in maintaining homeostasis, growth, development, metabolism, reproduction, and adaptation to changes in the internal and external environment.

Classification of Hormones

Hormones are mainly classified into three major groups based on their chemical nature:

- ***Peptide and Protein Hormones:*** These hormones are composed of amino acid chains and are generally water-soluble. They bind to receptors present on the surface of target cells and usually act through secondary messenger systems. Examples include insulin, growth hormone, and thyroid-stimulating hormone (TSH).

- ***Steroid Hormones:*** Steroid hormones are derived from cholesterol and are lipid-soluble. They can pass through cell membranes and bind to intracellular receptors, influencing gene expression and protein synthesis. Examples include cortisol, aldosterone, estrogen, and testosterone**.**

- ***Amino Acid-Derived Hormones:*** These hormones are synthesized from amino acids, mainly tyrosine and tryptophan. Their properties vary depending on their structure. Examples include epinephrine, norepinephrine, and thyroid hormones.

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29. What are the types of ion-exchange chromatography?

Types of Ion-Exchange Chromatography

Ion-exchange chromatography is an analytical separation technique based on the reversible exchange of ions between a charged stationary phase and ions present in the mobile phase. It is mainly used for the separation and purification of charged molecules such as proteins, peptides, amino acids, nucleotides, and inorganic ions. The separation occurs according to differences in the charge properties of molecules. Ion-exchange chromatography is mainly classified into two types:

1. Cation-Exchange Chromatography

In cation-exchange chromatography, the stationary phase contains negatively charged groups that bind positively charged ions (cations) from the sample. The negatively charged resin attracts and retains positively charged molecules, while uncharged or negatively charged molecules pass through the column. The bound cations can be released by changing the pH or increasing the concentration of salt in the mobile phase, which weakens the ionic interactions between the sample molecules and the stationary phase. Common cation-exchange resins contain acidic groups such as:

- Carboxyl groups (–COOH)
- Sulfonic acid groups ($–SO_3H$)

2. Anion-Exchange Chromatography

In anion-exchange chromatography, the stationary phase contains positively charged groups that bind negatively charged ions (anions) from the sample. Negatively charged molecules are attracted to the positively charged resin and remain attached, while other molecules move through the column. The bound anions can be eluted by altering the pH or increasing ionic strength of the mobile phase. Common anion-exchange resins contain basic groups such as:

- Amino groups ($–NH_2$)
- Diethylaminoethyl (DEAE) groups

30. Differentiate between DNA and RNA.

Difference Between DNA and RNA

DNA	RNA
DNA stands for Deoxyribonucleic Acid.	RNA stands for Ribonucleic Acid.
DNA contains the sugar deoxyribose in its nucleotide structure.	RNA contains the sugar ribose in its nucleotide structure.
DNA is usually double-stranded, forming a double helix structure.	RNA is usually single-stranded, although it may fold into complex structures.
The nitrogenous bases present in DNA are adenine (A), thymine (T), guanine (G), and cytosine (C).	The nitrogenous bases present in RNA are adenine (A), uracil (U), guanine (G), and cytosine (C).
DNA contains thymine as a pyrimidine base.	RNA contains uracil instead of thymine.
DNA is mainly located in the nucleus of eukaryotic cells, with small amounts present in mitochondria and chloroplasts.	RNA is found in the nucleus and cytoplasm, particularly in ribosomes.
DNA is more chemically stable due to its double-stranded structure and deoxyribose sugar.	RNA is less stable because of the presence of ribose sugar and its single-stranded nature.

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DNA	RNA
The main function of DNA is the storage and transmission of genetic information.	The main function of RNA is gene expression and protein synthesis.
DNA replication produces an identical copy of DNA during cell division.	RNA is synthesized from DNA through the process of transcription.
DNA molecules are generally very large and contain extensive genetic information.	RNA molecules are generally smaller and have specific functional roles.
DNA is present as chromosomes that carry hereditary information.	RNA exists in different forms such as mRNA, tRNA, and rRNA, each performing specific roles in protein synthesis.

31. Write a note on Voltage-Gated Channel.

Voltage-Gated Channels

Voltage-gated channels are specialized membrane proteins that regulate the movement of ions across the cell membrane in response to changes in the electrical potential (voltage) across the membrane. They play a crucial role in generating and transmitting electrical signals in excitable cells such **as** neurons, muscle cells, and cardiac cells. These channels are composed of transmembrane proteins embedded in the lipid bilayer. They contain voltage-sensing regions that detect changes in membrane potential and respond by opening or closing the channel, allowing specific ions to pass through.

Mechanism of Voltage-Gated Channels

At the resting membrane potential, voltage-gated channels remain closed, preventing ion movement across the membrane. When the membrane potential changes due to stimulation, the voltage sensor within the channel detects this change and causes the channel to open. Once opened, ions move through the channel according to their electrochemical gradient:

- Sodium ions (Na^+) enter the cell, causing depolarization of the membrane.
- Potassium ions (K^+) leave the cell, causing repolarization of the membrane.
- Calcium ions (Ca^{2+}) enter certain cells and participate in processes such as muscle contraction and neurotransmitter release.

After activation, many voltage-gated channels undergo inactivation, temporarily preventing further ion flow until the membrane returns to its resting state.

Types of Voltage-Gated Channels

1. Voltage-Gated Sodium Channels (Na^+ Channels): These channels open rapidly in response to membrane depolarization and are responsible for the rising phase of action potentials in neurons and muscle cells.

2. Voltage-Gated Potassium Channels (K^+ Channels): These channels open more slowly and allow potassium ions to leave the cell, contributing to membrane repolarization and restoration of the resting membrane potential.

3. Voltage-Gated Calcium Channels (Ca^{2+} Channels): These channels allow calcium ions to enter cells and are involved in neurotransmitter release, muscle contraction, and cellular signaling.

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32. Explain the biological order and disorder in living things.

Living organisms are highly organized systems in which different levels of biological structures are arranged in a specific and coordinated manner. This organization is maintained through complex interactions between molecules, cells, tissues, organs, and organisms. At the same time, biological systems also experience disorder due to random molecular movements, energy changes, and environmental influences.

Biological Order

Biological order refers to the organized arrangement and precise coordination of biological structures and processes that allow living organisms to function efficiently. This order exists at different levels of biological organization, from molecules to complete organisms. The levels of biological order include:

- ***Molecular Organization:*** The smallest level of biological organization involves atoms and molecules arranged in specific structures. Biomolecules such as proteins, nucleic acids, carbohydrates, and lipids have highly organized arrangements that determine their functions.
- ***Cellular Organization:*** Cells represent the basic structural and functional units of life. Within cells, organelles such as the nucleus, mitochondria, and ribosomes are arranged in an organized manner to perform specialized functions.
- ***Tissue and Organ Organization:*** Similar cells are arranged together to form tissues, and different tissues combine to form organs. The coordinated arrangement of tissues allows organs to carry out specific physiological functions.
- ***Organismal Organization:*** Multiple organ systems work together in a coordinated manner to form a complete organism. This organization maintains survival, growth, reproduction, and internal balance.

Biological Disorder

Biological disorder refers to the loss of normal organization or regulation within a living system. It occurs when cellular, molecular, or physiological processes become disturbed, leading to abnormal structure or function. Disorder in living systems may occur due to:

- ***Genetic Changes:*** Alterations or mutations in DNA can disrupt the normal production and function of proteins, leading to abnormal cellular activities.
- ***Cellular Dysfunction:*** Damage to cellular components, such as membranes, mitochondria, or enzymes, can interfere with normal cellular processes.
- ***Imbalance in Biological Regulation:*** Failure of regulatory mechanisms such as hormonal control, immune responses, or metabolic pathways can disturb normal body functions.
- ***Environmental Effects:*** Factors such as toxins, radiation, infections, and nutritional deficiencies can disturb biological organization and cause disease.

33. Explain all four structures of protein.

Four Levels of Protein Structure

Proteins are complex biological macromolecules composed of amino acids linked together by peptide bonds. The function and activity of a protein depend on its specific three-dimensional structure, which is organized into four levels: primary, secondary, tertiary, and quaternary structures.

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1. Primary Structure of Protein

The primary structure refers to the specific linear sequence and arrangement of amino acids in a polypeptide chain. Amino acids are connected through peptide bonds formed between the carboxyl group of one amino acid and the amino group of another. The sequence of amino acids is determined by the genetic information encoded in DNA. Even a single change in the amino acid sequence can affect the structure and function of a protein. The primary structure determines the higher levels of protein folding because the amino acid sequence influences the interactions that occur within the protein molecule.

2. Secondary Structure of Protein

The secondary structure refers to the local folding patterns of the polypeptide backbone caused mainly by hydrogen bonding between the carbonyl oxygen and amino hydrogen groups of peptide bonds. The two major types of secondary structures are:

a) Alpha (α) Helix

An alpha helix is a spiral arrangement of the polypeptide chain stabilized by hydrogen bonds formed within the same chain. The structure is usually tightly coiled and provides stability to the protein.

b) Beta (β) Pleated Sheet

A beta pleated sheet is formed when segments of a polypeptide chain align parallel or antiparallel to each other and are stabilized by hydrogen bonds between adjacent strands.

3. Tertiary Structure of Protein

The tertiary structure refers to the complete three-dimensional folding of a single polypeptide chain. It results from interactions between the side chains (R-groups) of amino acids. The forces responsible for maintaining tertiary structure include:

- Hydrogen bonding
- Ionic interactions
- Hydrophobic interactions
- Van der Waals forces
- Disulfide bonds between cysteine residues

The tertiary structure creates specific functional regions, such as active sites in enzymes and binding sites in proteins. It determines the biological activity of many proteins.

4. Quaternary Structure of Protein

The quaternary structure refers to the arrangement and interaction of two or more polypeptide chains (subunits) to form a functional protein molecule. Each subunit may have its own primary, secondary, and tertiary structures, and multiple subunits combine through various interactions to form the complete protein.
The forces maintaining quaternary structure include:

- Hydrogen bonds
- Ionic interactions
- Hydrophobic interactions
- Disulfide bonds

Proteins with quaternary structures show cooperative interactions between subunits, allowing efficient regulation of biological functions. This level of organization is present in many complex proteins composed of multiple subunits.

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